

changes to their products and operations use carbon credits as an escape route. This is known as greenwashing and undermines the credibility of the entire carbon market. Say, a company uses large amounts of fossil fuels. The damage they do to the environment is very high and long-lasting, but they get off easily if they have the money to spare. Including penalties and fines, the purchase of carbon credits often proves cheaper for them than revamping their processes to use sustainable energy sources. Hence, the trading of carbon credits needs to be very carefully planned and monitored by governments to make sure it does not delay their journey to net-zero.

- **Voluntary carbon offsets:** At this juncture, it is also important to talk about carbon offsets, a term used frequently alongside carbon credits. Carbon offsets are bought and sold in a voluntary market, where the players have no legal obligation to do so. It is like a voluntary investment to offset a company's carbon emissions, over and above what is mandated, or in countries where there is no legal framework in place yet. Companies like Microsoft, Google, Apple, and Amazon voluntarily invest in several such carbon offsetting projects, to accelerate their journey to net-zero or carbon negative status.

There are several global agencies that liaison between the projects/entities that generate carbon credits and offsets, and those willing to buy them. Apart from the government agencies for compliance trading, there are third party platforms where you can buy offsets, such as the United Nations Carbon Offset Program, Regreener, South Pole, Rabo Carbon Bank, ClimatePartner, Cloverly, Carbonplace, Amazon Sustainability Exchange, and Carbon Registry - India, among others. Although voluntary offset trading is



Soil carbon sequestration is one of the nature-based carbon removal technologies that tech companies are investing in, to offset their emissions (Courtesy: climatetransform.com)

quite low now, compared to regulatory carbon trading, it is expected to go up in the future as governments tighten their regulations. It is interesting to see private investors buying carbon offsets, akin to investing in stocks and other trade instruments! They are also putting a lot of money into carbon dioxide removal technologies.

Ideas from the trendsetters

Even though information technology and telecommunications do not currently feature among the top GHG emitting industries, tech majors across the globe and in India have taken the net-zero challenge quite seriously. And it is a great thing, because with the rise of AI, and deepening concerns about the power consumed by data centres and the amount of water required to keep the GPUs cool, it is good to proactively manage their carbon footprint right from now. Here's a quick preview...

Apple: Apple has an ambitious target of becoming carbon neutral across its supply chain by 2030. It has reportedly surpassed a 60% reduction in GHG emissions compared to 2015 levels, and has made significant strides in the use of renewable energy, waste management, and water conservation across its supply chain. The company and its suppliers aim to remove 9.6 million metric tons of carbon from the atmosphere annually through carbon removal projects by 2030.

The company is increasingly using recycled and renewable materials in its products. As of early 2025, Apple was 99% closer to its goal of using 100% recycled rare earth elements in all magnets and 100% recycled cobalt in all Apple-designed batteries.

It is also implementing strategies to reduce product emissions, such as designing Mac minis with recycled content and renewable electricity, and shipping them via low-carbon methods like ocean freight. The company is also tackling the problem of potent fluorinated gases used in semiconductor and display manufacturing, with a commitment from suppliers to reduce these emissions by at least 90%.

Apple also continues to be committed to its Restore Fund, in association with Conservation International and Goldman Sachs, which encourages members to profitably invest in natural carbon reduction and removal technologies such as forest restoration and mangrove conservation.

Microsoft: It is perhaps the only tech company that features in the league of top carbon credit buyers otherwise dominated by the energy sector. That is because the company not only aims to become carbon negative by 2030, it also hopes to compensate for all its emissions since inception by 2050! To achieve carbon negativity, the company is drastically reducing its emissions, procuring vast amounts of carbon-free renewable energy, electrifying its fleet, and investing in carbon dioxide removal technologies and nature-based solutions through its Climate Innovation Fund.

Microsoft uses mechanisms like internal carbon tax to weave the net-zero culture into every cost-centre. To offset unavoidable emissions, the